

Experiment 1 — Cloud Account Setup & Monthly Cost Estimation

Step 1: Cloud Account Creation

1. Go to <https://aws.amazon.com> and sign up for a free-tier account
2. Complete email, identity, and payment verification (card is required but won't be charged under free tier)
3. Log in to the **AWS Management Console**
4. Make sure your region is set to **Asia Pacific (Mumbai) — ap-south-1**

 **Screenshot needed:** AWS Management Console home dashboard showing your name/account in the top right

Step 2: Free Tier vs Paid Tier Table

Navigate to **Billing & Cost Management** → **Free Tier** to check your usage. Then note down a comparison table like this in your journal:

Free Plan	Paid Plan
Sign-up credits provided	Sign-up credits provided
Access to Always Free services	Access to Always Free + short-term trials
Access to select AWS services	Access to all AWS services
No charges during usage	Pay for charges exceeding credit balance
Account closes when credits deplete	Account stays active after credits deplete
Not eligible for promotional discounts	Eligible for promotional discounts

 **Screenshot needed:** Billing dashboard or Free Tier usage page

Step 3: Define the Cloud Workload

You'll be designing a three-tier architecture using **EC2 + S3 + RDS**. Navigate to each service and configure as follows:

EC2 (Compute):

1. Go to **EC2** → **Launch Instance**
2. Name it (e.g., your name + server)
3. Select **Amazon Linux 2023** AMI (Free Tier eligible)
4. Instance type: **t3.micro**
5. Create a new key pair, download the **.pem** file
6. Don't actually launch — just reach the summary screen

 **Screenshot needed:** EC2 launch summary panel showing AMI, instance type, key pair, storage

S3 (Storage):

1. Go to **Amazon S3** → **Create Bucket**
2. Enter a globally unique bucket name
3. Region: ap-south-1
4. Keep Block Public Access enabled
5. Don't actually create yet — just reach the config screen

 **Screenshot needed:** S3 bucket creation config screen

RDS (Database):

1. Go to **Amazon RDS** → **Create Database**
2. Choose **Easy Create**, engine: **MySQL** or **PostgreSQL**
3. Template: **Free Tier**
4. Note the instance class and storage shown

 **Screenshot needed:** RDS Create Database screen showing engine and free tier selection

The architecture for this three-tier setup follows a standard AWS VPC diagram (Application Load Balancer → EC2 instances → RDS, with S3 for static content). No further elaboration needed here.

Step 4: Cost Estimation Using AWS Pricing Calculator

Go to <https://calculator.aws/pricing/2/home>

EC2 Estimate:

1. Click **Add Service** → **EC2**
2. Region: US East (or Mumbai)
3. Instance: **t3.micro**, 1 instance, 730 hrs/month
4. OS: Linux
5. Tenancy: Shared

6. Note the monthly estimate (~\$18–19/month)

 **Screenshot needed:** Pricing calculator summary showing EC2 monthly cost

S3 Estimate:

1. Add another service → **S3**
2. Storage: ~10 GB Standard
3. Requests: minimal
4. Note the monthly estimate (~\$0.26/month)

 **Screenshot needed:** Pricing calculator summary showing S3 monthly cost

RDS Estimate:

1. Add another service → **RDS**
2. Engine: MySQL/PostgreSQL, db.t3.micro or similar
3. Storage: 100 GB
4. Note the monthly estimate — **it will be very high (~\$6,000+)**; this is intentional and part of the analysis

 **Screenshot needed:** Pricing calculator summary showing RDS monthly cost

Final Summary:

- Open AWS Console
- Billing and Cost Management to check usage
- Create EC2, S3, RDS and take screenshots
- Follow same in Pricing Calculator